SESSION 7 – Conditional Probability & Bayes' Theorem

1. Given the following data from an e-commerce app: out of 500 users, 120 bought phone accessories, 80 bought a phone, and 50 bought both, calculate the probability that a user bought a phone given they already bought accessories (P(Phone|Accessories)).

Answer:

Given:

- Users who bought accessories = 120
- Users who bought both phone and accessories = 50

Formula:

P(Phone | Accessories) = P(Phone and Accessories) / P(Accessories)

Using the given data:

P(Phone | Accessories) = 50 / 120

= 0.4167

= 41.67%

Therefore, the probability that a user bought a phone given that they already bought accessories is **41.67%**.

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2. Imagine you are analyzing Flipkart's user data: 30% of users buy headphones, and among those, 40% later buy a smartphone. Calculate the conditional probability that a randomly selected user buys a smartphone, given they have already bought headphones.

Answer:

Given:

- P(Headphones) = 30%
- P(Smartphone | Headphones) = 40%

The required conditional probability is:

P(Smartphone | Headphones) = 40%

Therefore, the probability that a user buys a smartphone given that they have already bought headphones is **40%**.

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3. Write a Python function `bayes_churn_probability(prior_churn, prob_email_given_churn, prob_email)` that uses Bayes' Theorem to compute the probability a user will churn given they received a "we miss you" email. Test your function with prior_churn = 0.15, prob_email_given_churn = 0.6, prob_email = 0.25.

Answer:

Bayes' Theorem:

P(Churn | Email) = [P(Email | Churn) × P(Churn)] / P(Email)

Given:

- P(Churn) = 0.15
- P(Email | Churn) = 0.60
- P(Email) = 0.25

Calculation:

P(Churn | Email) = (0.60 × 0.15) / 0.25

= 0.09 / 0.25

= 0.36

= 36%

Therefore, the probability that the user will churn given that they received the "we miss you" email is **36%**.

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4. A digital wallet app detects fraud based on spending patterns. If 2% of all transactions are fraudulent, and the app's alert system correctly flags 90% of fraudulent transactions but also falsely flags 5% of normal transactions, use Bayes' Theorem to calculate the probability that a transaction is actually fraudulent when flagged by the alert system.

Answer:

Let:

- F = Fraud
- A = Alert

Given:

- P(F) = 0.02
- P(A | F) = 0.90
- P(A | Normal) = 0.05
- P(Normal) = 1 - 0.02 = 0.98

First, calculate P(A):

P(A) = P(A | F) × P(F) + P(A | Normal) × P(Normal)

= (0.90 × 0.02) + (0.05 × 0.98)

= 0.018 + 0.049

= 0.067

Now apply Bayes' Theorem:

P(F | A) = [P(A | F) × P(F)] / P(A)

= (0.90 × 0.02) / 0.067

= 0.018 / 0.067

= 0.2687

= 26.87%

Therefore, the probability that a transaction is actually fraudulent when it is flagged by the alert system is approximately **26.87%**.

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5. Use ChatGPT or another AI tool to generate a real-world scenario where conditional probability is used in a food delivery app (like Zomato or Swiggy), and explain how Bayes' Theorem could help improve their recommendation system. Paste the AI's output and add one sentence of your own analysis.

Answer:

AI-generated output:

A food delivery app can use conditional probability to understand a customer's food preferences. For example, if a customer frequently orders South Indian food, the app can calculate the probability that the customer will order dosa given that they have previously ordered South Indian food.

For example:

P(Dosa | South Indian Food) = 70%

This means that when the app knows a customer prefers South Indian food, there is a 70% probability that the customer may order dosa.

Bayes' Theorem can help the recommendation system update its predictions when new information becomes available. For example, if a customer starts ordering more pizzas, the app can update the probability that the customer prefers pizza and show more pizza restaurants or offers.

My Analysis: Bayes' Theorem can help food delivery apps make recommendations more personalized by continuously updating customer preferences based on their previous orders.